Nathan Gardner Hattersley

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Summary

Ph.D. candidate in Economics at the University of Texas at Austin with an M.S. in Statistics. Specialize in causal inference, pricing, demand estimation, and structural industrial organization using large observational datasets. Experience spans antitrust litigation, global policy research, and production software engineering.

Education

- **University of Texas at Austin** — Austin, TX
  Ph.D. Economics (expected 2026); M.A. Economics — *Aug 2020–Present*
- **University of Arizona** — Tucson, AZ
  M.S. Statistics; B.A. Mathematics and Computer Science — *Aug 2013–May 2018*

Skills

- **Programming**: Python, R, Julia, SQL, TypeScript, Rust, Java, C, Stata, MATLAB, Bash
- **Economics & Statistics**: Causal inference, structural modeling, demand estimation, nonparametric methods, numerical optimization
- **Tools**: Pandas, tidyverse, Postgres/PostGIS, Docker, Git, HPC, Make/Snakemake, LaTeX
- **Languages**: English (native); Spanish (proficient)

Professional Experience

- **Texas Office of the Attorney General** — Austin, TX
  Economics Fellow — *Jul 2024–Jun 2025*
  - Analyzed antitrust cases involving online advertising auctions and algorithmic collusion using industrial organization frameworks.
  - Drafted expert rebuttals and economic analyses supporting litigation strategy alongside senior economists.
  - Identified a security vulnerability affecting a pivotal case, informing defensive legal strategy.
  - Communicated technical economic findings to attorneys and other non-technical stakeholders.
- **World Bank Group** — Remote
  Short-Term Consultant — *Feb 2021–Aug 2021*
  - Built a PCA-based index measuring political participation in Afghanistan using large-scale survey data.
  - Designed an algorithm incorporating ordinal survey rankings into composite participation measures.
  - Authored technical documentation and presented methodology and findings to project stakeholders.
  - Developed and delivered internal training on R, tidy data principles, and functional programming.
- **JPMorgan Chase & Co.** — Houston, TX
  Analyst (Software Engineering) — *Jun 2018–Jun 2020*
  - Developed and maintained production trading systems in Python and TypeScript supporting interest rate derivatives.
  - Designed regulatory workflows and trade models for new financial instruments.
  - First team migrating legacy platform to a microservice architecture using React, GraphQL, and WebSockets.
  - Collaborated with technical and business stakeholders to refine requirements, beta test software, and resolve deployment bottlenecks.

Research

- **Automobile Pricing**: Causal analysis of automobile dealership consolidation using transaction-level data to estimate effects on prices and consumer outcomes.
- **Dynamic Pricing**: Structural model of allocation incentives and pricing behavior under dynamic competition.
- **Consumer Demand**: Nonparametric estimation of grocery demand and pass-through using high-dimensional scanner data.

Additional Experience

- **Teaching**: Teaching assistant for Master's Econometrics, Industrial Organization, and Causal Inference. Led weekly review sessions, designed programming assignments, and taught statistical computing.
- **Research Computing**: Led workshops on high-performance computing for economics Ph.D. students and developed departmental guidance for parallel computing in Julia, Python, and MATLAB.